

Predicting Relative Article Popularity Using Multimodal Transfer Learning

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Abstract

We predict, at the moment of publication, whether a mako.co.il / n12.co.il news article will land in the top 10% of page views (first 48 hours) among articles on the same site in the same month, using only pre-publication signals: headline, thumbnail image, and site. We build four non-deep baselines (TF-IDF+LogReg, image-only, naively-concatenated title+image, and a fine-tuned Hebrew BERT text model) and two multimodal deep models: (1) staged fine-tuning of SigLIP2 (frozen head → partial unfreeze → LoRA), later extended with a cross-attention fusion head, and (2) a hybrid architecture combining AlephBERT (Hebrew-specific text encoder) with a ResNet50 image encoder. SigLIP2, in all tested configurations, did not surpass the text-only BERT baseline. The hybrid model, after tuning regularization and early-stopping granularity, became the only model in the project to exceed the text-only baseline's PR-AUC on the validation set (0.240 vs. 0.212). On the held-out test set this advantage narrows substantially (0.212 vs. 0.212), while ranking quality (ROC-AUC) is stable or slightly improved. We report this honestly as a case study in how aggressive validation-driven model selection across many iterations can inflate the apparent gap between models, and conclude that the image thumbnail provides at most a modest, non-conclusive improvement over the headline alone for this task.

1. Introduction & Related Work

1.1 Problem Statement

News organizations routinely need to judge, before publication, whether a given headline and thumbnail combination is likely to perform well. This project frames that judgment as a supervised binary classification problem: given an article's headline, thumbnail image, and publishing site, predict whether it will fall in the top 10% of page views (by first-48-hour traffic) among articles on the *same site* in the *same month*. The relative, peer-group-normalized target avoids conflating “popular” with “high absolute traffic,” which would otherwise be dominated by site-level and seasonal effects unrelated to editorial quality.

The central research question is whether the thumbnail image adds predictive value beyond the headline text alone. This is not obvious a priori: headlines are a strong, information-dense signal (an editor's distilled pitch for the piece), while thumbnails on Israeli news sites are frequently generic stock or wire-service images only loosely tied to article content. Answering this question required not just training one multimodal model, but *isolating* the image's marginal contribution through a series of controlled ablations — which is the organizing principle of this report.

1.2 Related Work

- Tschannen et al., “SigLIP 2: Multilingual Vision-Language Encoders with Improved Semantic Understanding, Localization, and Dense Features,” arXiv:2502.14786, 2025. Provides the pretrained multilingual vision-language encoder used as the primary multimodal backbone.
- Abousaleh, Cheng, Yu, and Tsao, “Multimodal Deep Learning Framework for Image Popularity Prediction on Social Media,” IEEE Transactions on Cognitive and Developmental Systems, vol. 13, no. 3, 2021. Directly motivates fusing visual and contextual (textual) signals for popularity prediction, over text-only or image-only models.

- Bandari, Asur, and Huberman (and the broader “Online News Popularity” literature it spawned), “Predicting the Popularity of Online News from Content Metadata,” IEEE, 2017 — predicts pre-publication popularity (shares/likes/comments) on the widely-used Mashable benchmark using gradient-boosted trees over content-metadata features (keyword statistics, day of week, word counts). Directly comparable in framing (before-publication-only features) but, like most of this literature, restricted to text/metadata features — it does not use the article's actual image content.
- “News Popularity Prediction with Machine Learning,” a comparative study of classical models (Random Forest, Bayes Net, C4.5) on keyword and content-metadata features, with Random Forest + PCA performing best by ROC. Confirms the same metadata-only pattern: our use of the raw thumbnail image itself, rather than text-derived metadata about it, is a point of difference from this literature, not something we found direct precedent for.

A preliminary zero-shot diagnostic (no fine-tuning, ~950 real headline-image pairs) found SigLIP2's top-1 image-retrieval accuracy from Hebrew titles to be roughly 28x the random baseline — evidence that the pretrained backbone carries *some* cross-modal signal in Hebrew despite being English-heavy in its training mix, which justified committing to full fine-tuning rather than abandoning the multimodal approach at the outset.

2. Methodology (Math & Architecture)

2.1 Data

Source: Keshet's internal Snowflake page-view event warehouse, covering mako.co.il and n12.co.il. For each article we extract the headline (`teaser_title`), thumbnail URL (`pic_furl`), site, and publication timestamp, plus first-48h page views (label construction only, never a model input). Formally, for site s and calendar month m , an article i is labeled positive iff its 48-hour page-view count v_i satisfies

$$v_i \geq Q_{0.9}(\{v_j : \text{site}(j)=s, \text{month}(j)=m\})$$

i.e. it is at or above the 90th percentile of its own site-month peer group.

Extraction and split. 97,309 rows from January 2024 onward. A **time-based** (not random) split avoids leakage from near-duplicate or related articles across the train/val boundary:

Split	Period	Rows	Positive rate
Train	2024-01 – 2026-04	87,334	~10.0%
Val	2026-05	3,722	~10.0%
Test	2026-06	3,745	~10.0%

Images. Thumbnails are fetched by URL; 71,536/71,546 unique URLs (99.99%) downloaded successfully. A significant data-quality finding during this process: the URL as stored in the warehouse (`pic_furl`) by default resolves to a tiny 82x62px CDN thumbnail, not the original image. Stripping the CDN suffix recovers the full-resolution original when it still exists; a `..._autoOrient_b.jpg` variant (232x175px) serves as a reliable fallback. Final quality distribution (2,000-row sample): 60.8% full-resolution, 35.1% the 232x175 fallback, 4.1% stuck on the 82x62 thumbnail. This resolution variance was later used as a controlled experimental variable (Section 3.3).

Design decision: no tags as a core input. 49.5% of articles carry no tags at all, so tags were excluded from the core model (headline + image + site only) and treated as a secondary enrichment question that was not pursued further given the project timeline.

2.2 Baseline Models

Four models isolate the contribution of each input modality individually:

- 1. `tfidf_logreg` — TF-IDF over headline text + logistic regression.
- 2. `text_only_bert` — AlephBERT (Hebrew BERT) fine-tuned end-to-end on headline text.
- 3. `image_only` — frozen SigLIP2 image embeddings + logistic regression.
- 4. `title_image_frozen` — frozen SigLIP2 text and image embeddings, naively concatenated, into one logistic regression. Tests whether *naive* (non-jointly-trained) fusion of the two modalities helps at all.

2.3 Model 1: SigLIP2 Staged Fine-Tuning

`SigLIP2Classifier` (`google/siglip2-base-patch16-224`) encodes headline and thumbnail jointly, adds a learned 8-dim site embedding, and a small MLP head. Fine-tuning proceeds in three increasingly aggressive stages:

- **Stage A** — backbone fully frozen; only the head (395K params, 0.11% of the model) trains.
- **Stage B** — top 4 transformer layers unfrozen (57.1M params, 15.2%), with a discriminative learning rate (head at full LR, unfrozen backbone at LR × 0.1).
- **Stage C** — LoRA (r=8) on `q_proj/v_proj` across all 12 layers.

LoRA formulation. For a pretrained weight matrix $W_0 \in \mathbb{R}^{(d \times k)}$ (here, the attention projection matrices), LoRA freezes W_0 and represents the fine-tuning update as a low-rank decomposition:

$$W = W_0 + \Delta W = W_0 + BA, \text{ where } B \in \mathbb{R}^{(d \times r)}, A \in \mathbb{R}^{(r \times k)}, r \ll \min(d, k)$$

so the forward pass becomes $h = W_0x + BAx$. Only A and B are trained (r=8 here), giving ~986K trainable parameters versus 57.1M for full-layer unfreezing at comparable adaptation capacity — the parameter-efficiency that motivates LoRA’s use as this project’s “Advanced Training Strategy” component alongside standard transfer learning.

After all three stages under-performed `text_only_bert`, two further variants tested whether the fusion mechanism itself (naive concatenation of pooled embeddings) was the bottleneck:

- **Model 1** (exp04) — Stage B + Stage C combined, and a cross-attention fusion head in place of concatenation, plus full-resolution-only images.
- **Model 1b** (exp05) — a controlled follow-up isolating the fusion variable alone: identical capacity to Stage C (LoRA only), on the full dataset.

Cross-attention formulation. Standard scaled dot-product attention:

$$\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k}) \cdot V$$

implemented bidirectionally via `nn.MultiheadAttention`: the pooled text representation serves as Q against the image’s raw patch tokens as K=V (196 tokens from `vision_model.last_hidden_state`), and symmetrically the pooled image representation serves as Q against the raw text tokens as K=V — letting text “query” specific image

regions and vice versa, rather than only comparing pre-summarized vectors.

2.4 Model 2: Hybrid (AlephBERT + ResNet50)

A structurally different architecture: two backbones **never jointly pretrained** (unlike SigLIP2’s contrastively-aligned towers), fused only at the classification head.

- **Text:** AlephBERT (Hebrew-specific), CLS-token pooling.
- **Image:** ResNet50 (ImageNet-pretrained), global-average-pooled 2048-dim features.
- **Fusion:** concatenation + a small MLP — motivated directly by Section 3.3’s finding that cross-attention did not help once isolated from confounds, and by the absence of a shared pretrained embedding space here to begin with.

Both encoders are frozen by default, with a configurable number of top layers/blocks unfrozen, using the same discriminative-LR pattern as Model 1.

2.5 Training Objective and Metrics

Loss. All models optimize class-weighted binary cross-entropy:

$$L = -(1/N) \cdot \sum_{i=1}^N [w_+ \cdot y_i \cdot \log \sigma(z_i) + (1-y_i) \cdot \log(1-\sigma(z_i))]$$

with positive-class weight $w_+ = 9.0$ (matching the ~1:9 class imbalance), logit z_i , and sigmoid σ . Optimizer: AdamW with a cosine learning-rate schedule; early stopping on validation PR-AUC.

Metrics. With precision = TP/(TP+FP) and recall = TP/(TP+FN), **PR-AUC** is the area under the precision-recall curve swept over all thresholds, and **ROC-AUC** is the area under the true-positive-rate vs. false-positive-rate curve — equivalently, the probability that a randomly chosen positive example is scored higher than a randomly chosen negative one. Under class imbalance, a random classifier scores ROC-AUC = 0.5 but PR-AUC ≈ the base rate (≈ 0.10 here) — so PR-AUC, not accuracy or ROC-AUC alone, is the primary metric throughout, alongside F1 and Precision@K (K ∈ {50, 100, 200}), the latter matching the intended editorial ranking use case.

All experiment hyperparameters live in one YAML config per run (`configs/exp01-exp08`) for reproducibility; training ran locally on Apple Silicon (MPS backend).

3. Experiments & Results

3.1 Baselines

Model	Inputs	ROC-AUC	PR-AUC	F1
<code>tfidf_logreg</code>	title	0.660	0.208	0.245
<code>text_only_bert</code>	title (+tags)	0.704	0.212	0.248
<code>image_only</code>	image	0.619	0.148	0.229
<code>title_image_frozen</code>	title+image (naive concat)	0.619	0.147	0.214

The text-only BERT baseline is the strongest of the four by a clear margin. Critically, `title_image_frozen` **is no better than `image_only` alone, and worse than `text_only_bert` alone** — naively concatenating frozen embeddings from two modalities does not combine their signal.

3.2 SigLIP2 Staged Fine-Tuning

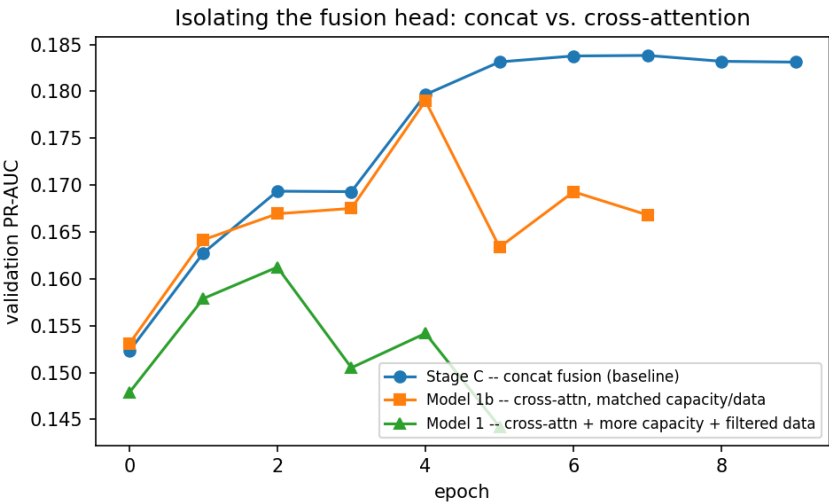
Stage	Trainable params	ROC- AUC	PR-AUC	Outcome
A (frozen)	395.8K (0.11%)	0.651	0.167	Improves over naive baselines, still below text_only_bert
B (unfreeze top 4)	57.1M (15.2%)	—	—	Overfit almost immediately; did not beat Stage A
C (LoRA, r=8)	986K (0.28%)	0.659	0.184	Best of the three stages; stable across all 10 epochs

None of the three staged variants surpassed text_only_bert (0.704 / 0.212). Stage C's stability confirms the expected pattern: on ~87K training rows, 57M trainable parameters overfit quickly, while LoRA's ~1M parameters generalize far better for comparable adaptation capacity.

3.3 Isolating the Fusion Head: Cross-Attention

Run	Fusion	Capacity	Data	Best ROC- AUC	Best PR- AUC
Stage C	concat	LoRA only	full (87.3K)	0.659	0.184
Model 1 (exp04)	cross- attention	unfreeze+LoRA combined	full-res filtered (53.5K)	0.625	0.161
Model 1b (exp05)	cross- attention	LoRA only (matched to Stage C)	full (87.3K)	0.643	0.179

Model 1 changed three variables simultaneously and regressed sharply. Model 1b isolates the fusion variable alone: the result (0.643/0.179) is close to but consistently below Stage C's concat-fusion result (0.659/0.184) across the whole run. **Conclusion: cross-attention fusion, as implemented here, was not the fix — Model 1's regression was driven mainly by added capacity and reduced data, not the fusion mechanism.**

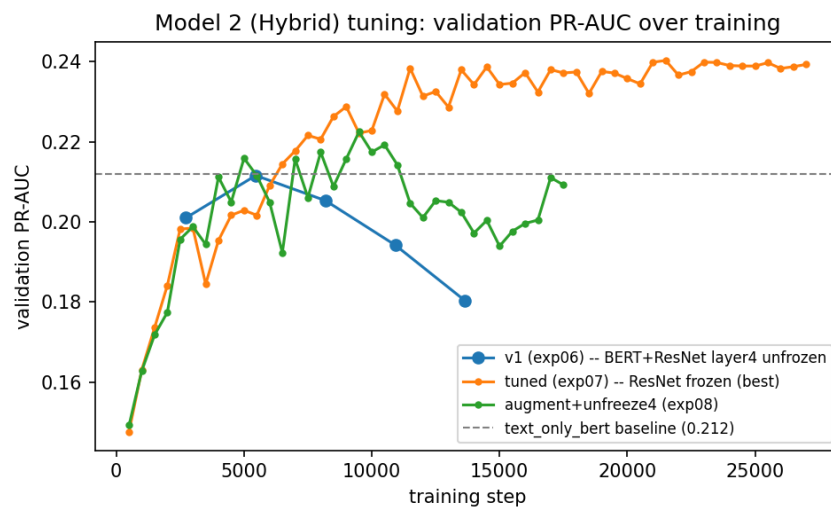


Isolating the fusion head: concat (Stage C) stays consistently above cross-attention at matched capacity (Model 1b); Model 1's added capacity and filtered data cause a collapse after epoch 4.

3.4 Hybrid Model (Model 2)

Run	Config	Best ROC-AUC	Best PR-AUC	Outcome
v1 (exp06)	BERT top-1-layer + ResNet layer4 unfrozen, concat fusion	0.660	0.212	Peaked epoch 1, overfit steadily afterward
tuned, first attempt	ResNet fully frozen, finer eval	0.670	0.198	Early-stopped by a patience/eval-frequency scaling bug – false stop
tuned (exp07)	same, bug fixed	0.693	0.240	Completed all 10 epochs with no early stopping; best result
augment+unfreeze4 (exp08)	exp07 + re-unfrozen ResNet + augmentation	0.674	0.222	Beat v1 but did not beat exp07

exp07 is the only model in this project to exceed text_only_bert on PR-AUC (0.240 vs. 0.212). Two design choices were decisive: (1) a Hebrew-specific text encoder (AlephBERT) rather than SigLIP2’s multilingual encoder, and (2) conservative regularization — ResNet50 fully frozen, higher weight decay, and early-stopping patience correctly scaled to the finer evaluation frequency.



Model 2 tuning journey: v1 overfits and degrades after ~step 5000; exp07 (ResNet frozen) climbs steadily past the text-only baseline with no collapse; exp08 improves on v1 but plateaus below exp07.

3.5 Final Ablation Table

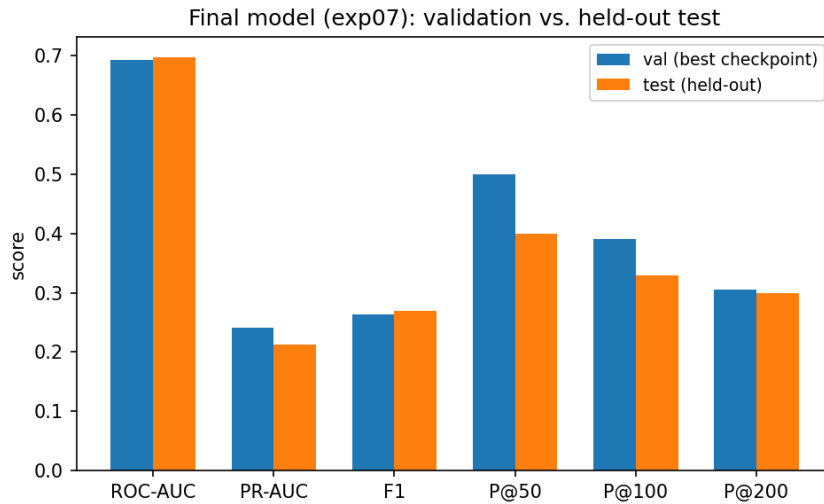
Ablation entry	Inputs	ROC-AUC	PR-AUC
title_only	title	0.660	0.208
title_tags*	title (no tags baseline exists)	0.704	0.212
image_only	image	0.619	0.148
title_image	title+image (naive concat)	0.619	0.147
full_model	title+image+site (exp07 , no tags)	0.693	0.240

*No model in this project was given tags as an input (Section 2.1); title_tags here is really title-only text, kept for the table’s canonical shape. The cleanest single comparison for “does the image help” is full_model vs. text_only_bert : 0.240 vs. 0.212 PR-AUC.

3.6 Held-Out Test Set

Split	ROC-AUC	PR-AUC	F1	P@50	P@100	P@200
Val (best checkpoint, step 21,500)	0.693	0.240	0.263	0.50	0.39	0.305
Test (held-out)	0.697	0.212	0.270	0.40	0.33	0.30

ROC-AUC is stable (marginally higher on test). **PR-AUC drops from 0.240 to 0.212**, landing almost exactly at text_only_bert’s validation PR-AUC, erasing most of the apparent advantage seen during tuning — attributed to selection-driven overfitting to the validation set across exp06 → exp07 → exp08 (every tuning decision was made by watching the same ~3,700-row validation set).



Final model, validation vs. held-out test: ROC-AUC and F1 hold up, while PR-AUC and Precision@K drop toward the text-only baseline's level.

4. Discussion & Conclusion

Does the image help? Modestly, and not conclusively. On validation data the best multimodal model (exp07) beat the strongest text-only baseline by a sizeable margin; on a single untouched test month that margin nearly disappears on PR-AUC while holding up on ROC-AUC. Given the well-understood mechanism (repeated validation-based model selection), we read this as “the image’s contribution is real but small, and was likely overstated by how many tuning decisions were routed through the same validation set” rather than “the image adds nothing.”

Why did the more sophisticated fusion approach fail? The controlled ablation (Section 3.3) is fairly definitive for this implementation: it was the *added trainable capacity and reduced training data*, not the *mechanism* (cross-attention vs. concatenation), that hurt Model 1. A more thorough treatment might test cross-attention at matched capacity across a wider sweep, or full token-to-token attention rather than pooled queries — both left unexplored given the project’s time budget.

Why did the Hybrid model succeed where SigLIP2 didn’t? A genuinely Hebrew-specific text encoder instead of a lightly-Hebrew-exposed multilingual one, plus conservative regularization discovered through iteration. This suggests language-appropriateness of the text backbone mattered more, for this task and language, than the theoretical advantage of a jointly-pretrained, contrastively-aligned text/image embedding space.

Limitations. (1) A single val/test month means month-to-month metric volatility is not separately quantified. (2) Tags were excluded entirely; their marginal value was never measured. (3) Thumbnail quality varies and was only

controlled for in one experiment. (4) A single random seed per configuration; no variance estimate across seeds.

Conclusion. We built and rigorously compared six modeling approaches — four single-modality baselines, staged SigLIP2 fine-tuning across three capacity regimes with two fusion mechanisms, and a Hebrew-specific hybrid architecture — to answer whether a news article’s thumbnail image predicts its relative popularity beyond its headline. The best model (AlephBERT + frozen ResNet50) improved PR-AUC over a strong text-only baseline on validation data (0.240 vs. 0.212) but that advantage substantially narrowed on a genuinely held-out test month (0.212 vs. 0.212), while ranking quality (ROC-AUC) held up. We conclude the thumbnail carries a real but modest signal beyond the headline, and that naive fusion of frozen multimodal embeddings is actively counterproductive — joint, task-specific fine-tuning with an appropriately-matched (Hebrew-specific) text encoder is what allowed any image contribution to surface at all.

References

1. M. Tschanen et al., “SigLIP 2: Multilingual Vision-Language Encoders with Improved Semantic Understanding, Localization, and Dense Features,” arXiv:2502.14786, 2025.
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3. “Predicting the Popularity of Online News from Content Metadata,” IEEE Conference Publication, 2017.
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